

Project Design Phase

Problem - Solution Fit Template

Date	01 October 2026
Team ID	SWTID-2026-2051
Project Name	Script-Controlled ACL - Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Problem - Solution Fit Template:

The Problem-Solution Fit means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps identify behavioral patterns and recognize what would work and why.

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and marketing strategy with the right triggers and messaging.
- Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- **Understand the existing situation in order to improve it for your target group.**

Template:

1. CUSTOMER SEGMENT(S) [CS] • ServiceNow admins and developers managing institutional / student data • Branch staff and faculty (e.g., EEE) who need only their own branch's records • Institution admins responsible for data privacy	6. CUSTOMER CONSTRAINTS [CC] • Needs security_admin role to create and edit ACLs • Scripting knowledge required for advanced ACLs • Must not break access for admins or legitimate users • Limited time and no budget for extra tools	5. AVAILABLE SOLUTIONS [AS] • Basic role-based ACLs: table-level only, no field-value filtering • Manual checking of access • Separate tables / instances per branch: costly, hard to maintain • Pros: simple. Cons: inflexible, not scalable, no record-level control
2. JOBS-TO-BE-DONE / PROBLEMS [J&P;] • Restrict who can view, create, edit and delete records • EEE users see only EEE records; admins keep full access • Protect sensitive student data (email, phone, faculty details)	9. PROBLEM ROOT CAUSE [RC] • Table-level access does not separate records by field value (Branch) • Without record-level rules, anyone with table access sees all branches • Read, create, write and delete permissions are often not configured separately	7. BEHAVIOUR [BE] • Admins create custom roles and assign them to users • Test access by impersonating different users • Check list views to confirm what each user sees • Update ACL conditions when access rules change
3. TRIGGERS [TR] • A user sees another branch's student records • A data privacy or audit requirement is raised • A new table with sensitive data needs protection	10. YOUR SOLUTION [SL] • Script-controlled ACLs on the u_institution_details table • Read ACL (bb1): data condition 'Branch is EEE' plus advanced script (admin full access, bb1 sees EEE records, others denied) • Create (bb2), Write (bb3) and Delete (bb4) ACLs, each with its own role • Verified by impersonating the EEE user, a user without roles, and Admin • Result: record-level security without duplicating tables	8.1 ONLINE (CHANNELS OF BEHAVIOUR) [CH] • ServiceNow instance: Users, Roles, Tables and Access Control (ACL) modules • ServiceNow docs, community forums and online tutorials
4. EMOTIONS: BEFORE / AFTER [EM] • Before: worried about data leaks, unsure who can see or change records • After: confident, in control and secure; each role has exactly the access it needs		8.2 OFFLINE [CH] • Team and department-head discussions on who should access what • Written access policies and review meetings

References:

1. <https://www.idealhackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>